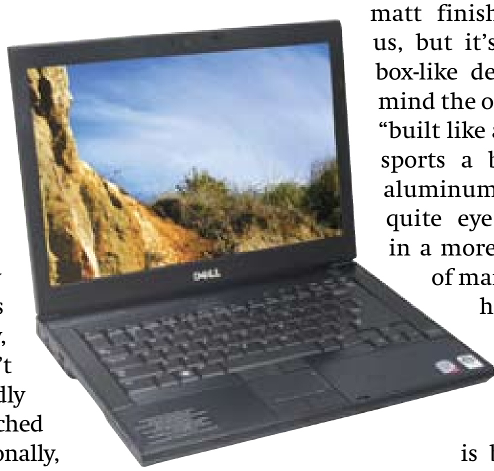
ally have an inflatable foam cladding around the hard drive that absorbs impacts thereby protecting the hard drive from crashes due to any sort of hard impacts. Business notebooks (in general) sport more rugged finishes that are less likely to wear with heavy usage. We call such choice of tough surface treatments *industrial grade*; this largely means something that's resilient to wear, although it's also become synonymous with a very plain-looking choice of materials. Most hardcore business notebooks pay a lot of attention to ergonomics; by this we mean both the presence and the placement of shortcut keys/buttons for common tasks.

This time round we decided to be really selective when looking at business-class notebooks. We didn't want our test to be diluted with pseudo-business notebooks. We did a more comprehensive test back in June last year and unfortunately all of the major players like HP, Dell, Lenovo, Sony, Toshiba and Fujitsu didn't have many new models. Sadly all these brands have launched a slew of models internationally, but most of them haven't released these models in India. In fact latter two didn't even have a single new model between them. Our five participants consisted of two notebooks from Lenovo and one each from HP, Sony and Dell.

Dell Latitude E6400

Brick solid!

If Dell's Vostro series is designed for the small to medium sized business on a budget, the Latitude series is designed for the serious corporate, who needs connectivity and reliability above all else. Think of the Latitude as Dell's answer to Lenovo's ThinkPad series. At first glance the Latitude E6400 looks very large, solid and also very boxy. It seems like Dell hewed this out of a solid hunk of plastic and metal. No, the black matt finish is fine with us, but it's just that the box-like design brings to mind the oft used analogy "built like a tank". The lid sports a black, brushed aluminum finish that is quite eye catching but in a more laid back sort of manner and you'll hardly notice yourself giving it another look over.



Dell Latitude E6400

The E6400 is built around a magnesium alloy cage which is great for longevity and even the coating around the palm rest region looks like it's there to last. In fact the mostly metal design does add to the weight a bit, but the E6400 feels surprisingly dense; as if it would take a lot of abuse. The screen size is 14.1-inches, although this notebook has a rather wide bezel all around the screen, which makes it look much larger, though not as large as a 15.4-inch model.

It sports the double mouse button design that Lenovo also favours and the track button that is a must for all seriously corporate notebooks. The track button isn't red however (Lenovo nee IBM patented that), but is black; and blends in with the keyboard. The keys themselves are well laid out and offer a short and very positive feedback. Although this wasn't the best laid out keypad from amongst the five notebooks we tested, the key spacing and beveling is ergonomic to work with. The trackpad is the right combination of grip and comfort although we found tracking to be a bit of a problem. In fact this was one of the major let downs with the E6400; the track pad will stick at times or will just refuse to do what your finger commands – most annoying. The trackpoint works better and it's clear to see that it isn't an adornment for Dell; many ThinkPad users swear by the trackpoint and Dell implements it quite well. A fingerprint reader thrown in; a must for many corporate users as an additional security step as it restricts access like few passwords can.

The screen itself is a matte panel; so reflections are not a problem and the resolution of 1440 x 900 pixels is absolutely perfect. We're tired of 1280 x 800 pixels and ultra high resolution notebooks are a no-no too. This resolution is just right for the screen size. The LCD panel is LED lit; and should bring power saving benefits as well. In terms of configurability the E 6400 is totally customisable in true Dell fashion. The one we got came with a new T9400 processor. This CPU is quite fast and runs at a speed of 2.5 GHz while maintaining a whopping 6 MB of L2 cache – great for someone needing more CPU performance. It seems the E6400 ships with only 7200 RPM hard drives and this is a very good thing because the storage subsystem is usually the slowest component of any PC or notebook. SSD is also an option with the new Latitude; though the costs are astronomical. In true workstation style the Latitude E6400 ships with an NVIDIA Quadro graphics solution; the NVS 160M which is hardly powerful, being based around eight stream processors but is way faster than Intel's GMA and about a tenth of the performance away from a GeForce 9300M GS. D-Sub and S-Video connects are provided as video outs.

The Latitude E 6400 is priced from Rs. 60,000 and above; although the model we recieved was priced at Rs. 95,000; not bad considering the processor and graphics solution.

HP Elitebook 6930P

All that glitters is not gold

The 6930P is yet another rather unobtrusive looking notebook. The lid itself is a matte black, unlike the machined metal finish on the Latitude E 6400 and looks even simpler; some might feel a little to plain looking. The body has an equally lack luster finish with dark grey and black used as tones. This sort of body finish is common to Lenovo's ThinkPads as well; yet the Elitebook looks even plainer. The culprit, we feel, is the boxy build which gives it a very (yawn) boring look. It's also about as heavy as the E6400. Once you open the lid, things get a little more interesting. The palm rest is silver-grey as is the outer bezel and lid hinges. The side bezel is slimmer than the E6400 although the top bezel is just as wide, making this notebook appear slightly bigger than its screen size of 14.1-inches; its footprint is a little smaller than the Dell Latitude E6400 though.

HP Elitebook 6930P

HP advertises that the Elitebook 6930P is designed with conformity to certain military standards, (MIL-STD 810F for those who want to look it up), for resilience to vibration, dust, humidity, altitude and high/low temperatures. This notebook also has the option to upgrade the optical drive to a special space saver ODD which allows for using an additional hard drive – a cool option for the power business user. RAID 0 and 1 support is also available if you decide to go dual hard drive. The drives themselves have an impact protection cage to minimize data crashes. The display is LED-backlit once again a plus for battery life; the quality of display remains as good as a CCFL display. The resolution is an excellent 1440 x 900 pixels, although you can opt for a screen with a resolution of 1280 x 800 pixels as an option.

The trackpad is small and quite recessed; a plus during heavy typing sessions as your wrist is less likely to come in contact with the touchpad surface thereby causing your mouse pointer to jump elsewhere which is a most irritating phenomenon. The touchpad also incorporates dual scroll zones for both vertical and horizontal scrolling. The keys on the 6930P sport the same kind of matte finish that the outer body sports and this adds a bit of welcome grip to each key. Key travel is a little more than the E 6400 and feedback is slightly softer. However the keys are well laid out and due to their large design lack beveling on the top. Needless to say this keypad worked well for all of us and although the theory this should not have been the case but in practice we could find little fault – we'd like key travel to be slightly less and feedback to be more on the positive side of responsive. There are a number of backlit and touch sensitive buttons on the top of the keypad and the all important WLAN on/off control is found here. The backlit icon changes colour when WLAN is switched on and off and there is another indicator which lights up on the front of the notebook; although this indicator will not be on eye level when you're using it so is not that useful. A configurable shortcut key and the volume buttons are also backlit and touch activated – this is a nifty addition to a business-class notebook; after all even corporate users like to indulge in a little soft finger touching! A fingerprint scan sensor is also provided to added security.

HP provides software that can manage and even repair your notebooks installation over a network; very useful for IT professionals and business houses looking for a robust fleet of notebooks with minimal down times. The 6930P is adequately powered

How We Tested

For features we looked at the bundle and the specifications of each laptop. We also looked at the connectivity options like number of USB ports, presence of FireWire, memory card readers and HDMI. We also looked for LAN, Bluetooth and WLAN connectivity.

We gave scores ranging on a scale of 10 for usability, intuitiveness, convenience and ergonomics too. Such non tangible factors are often what make the difference to most users. Features like build quality, or the feel of a keypad, or even the presence of and placement of shortcut keys. We also look at the ergonomics associated with the placement of USB ports, LAN and RJ11 connects and other connectivity options.

We used a clean install of Windows Vista Business 32 bit, and only the drivers and necessary benchmark software were installed. Our tests consisted of a gamut of benchmark suites and real world tests.

For checking the performance of key components like CPU, RAM, storage and video subsystems we used two general benchmarks: PC Mark 05, and SiSoft Sandra 2008 Lite. For specifically stressing the graphics subsystem we used 3D Mark 05. Maxon's CineBench R10 was used for testing the CPUs rendering ability. For testing the quality of the display provided we used DisplayMate's colour suite. Finally, we used WirelessMon 2.0 to check the strength of the WLAN solution embedded on each notebook. We also benchmarked the transfer rate of the WLAN system. We transferred a single 100 MB file for the sequential test and a number of small files totaling to 100 MB in size for the random test.

For real world tests, we checked the overall effectiveness of each of the test subjects at viewing HD content and gaming, music, file compression/decompression (WinRAR), and file transfers. The former tests stress out the video solution and CPU and memory subsystems while the file transfer test specifically targets the storage subsystem.